

Data Preprocessing

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Abstract

The project Smart Motortargets to create an AI algorithm where a motor controller can decide to increase/decrease motor speed based on trained time series data.

this report The analysis of the FordA dataset, a benchmark dataset with labeled motor signal data, is presented in this report. Assessing the quality and structure of the dataset and organizing suitable preprocessing procedures for model creation are the goals. The data-related contributions made during the project's early stages are the main topic of this report.

Keywords: *Time-series preprocessing, Dataset evaluation, Motor signal cleaning*

1. Material

The FordA dataset was used for this work. It is accessible to the general public via OpenML. The dataset includes:

- Each of the 3,601 samples has 500 time steps.
- One sensor signal is represented by each sample in univariate signal data. Labels for binary classes:
 - -1 = normal motor signal
 - 1 = abnormal motor signal

Each row in the tabular dataset represents a single signal, and each column represents a time step. The labels are utilized for supervised classification jobs and are kept apart. The following resources and technologies were used in this analysis:

- Python
- Pandas—for exploring data
- Matplotlib—for visualizing distributions and signals
- scikit-learn — for preparing and importing data

2. Method

2.1. Exploration and Dataset Loading

The sklearn.datasets module's `fetch_openml()` was used to load the dataset. To verify the quantity of samples, time steps, and class labels, an initial inspection was conducted. Pandas was used to examine the data in order to verify the sample values and shape.

2.2. Visualization of Data

Matplotlib.pyplot.plot() was used to plot the initial samples in order to visualize their waveform structure. To comprehend variance and possible features, normal and aberrant signals were visually examined.

2.3. Data Quality Checks

Among the important checks were:

- Missing Values: used `X.isnull().sum().sum()` to check
- Label Distribution: `y.value_counts()` was used to verify
- Signal Variability: Each sample's mean and standard deviation were determined
- Plotted to analyze the value distribution of individual signals, a histogram

3. Database concept

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graphic2.png

Figure 1. Example figure

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A sample equation 1, shows the Schrödinger equation as an example.

$$\frac{\hbar^2}{2m} \nabla^2 \Psi + V(\mathbf{r})\Psi = -i\hbar \frac{\partial \Psi}{\partial t} \quad (1)$$

The `amssymb` package was not necessary to include, because stix2 font incorporates mathematical symbols for writing quality equations. In case you choose another font, uncomment the package in `rho class`.

In case you want to change the values that adjust the spacing above and below in the equations, go to `rho-class/rho.cls/math packages` section and play with `\setlength{\eqskip}{8pt}` value until the preferred spacing is set.

4. Implementation aspects

4.1. Aspect number 1

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4.2. Aspect number 2

4.3. Inserting Code

This class includes the *listings* package, which offers customized features for adding codes specially for C, C++, $\text{\LaTeX}$ and Matlab. You can customize the format in *rho* class file.

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```

18         fib_sequence(i) = fib_sequence(i-1) +
19         fib_sequence(i-2);
20     end
21
22     fprintf('Fibonacci Sequence:\n');
23     disp(fib_sequence);
24 end

```

Code 1. Example of matlab code.

Insert code with caution. The code must be self explanatory or it must be explained in the text. Only show important parts of your code.

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5. Testing

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5.1. Insert a sample figure

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Here is how to insert graphics. Upload the file in Overleaf and use the following commands.

Figure 2 shows an example figure.

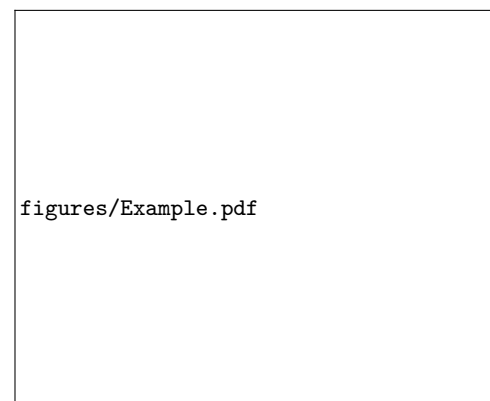


Figure 2. Example figure obtained from PGFPlots [4].

5.2. Sample double figure

Figure 3 shows an example of a two-picture floating figure that covers the width of the page. It can be positioned at the top or bottom of the

```

1 function fibonacci_sequence(num_terms)
2     % Initialize the first two terms of the
3     % sequence
4     fib_sequence = [0, 1];
5
6     if num_terms < 1
7         disp('Number of terms should be greater
8         than or equal to 1. ');
9         return;
10    elseif num_terms == 1
11        fprintf('Fibonacci Sequence:\n%d\n',
12        fib_sequence(1));
13        return;
14    elseif num_terms == 2
15        fprintf('Fibonacci Sequence:\n%d\n%d\n',
16        fib_sequence(1), fib_sequence(2));
17        return;
18    end
19
20    % Calculate and display the Fibonacci
21    % sequence
22    for i = 3:num_terms

```

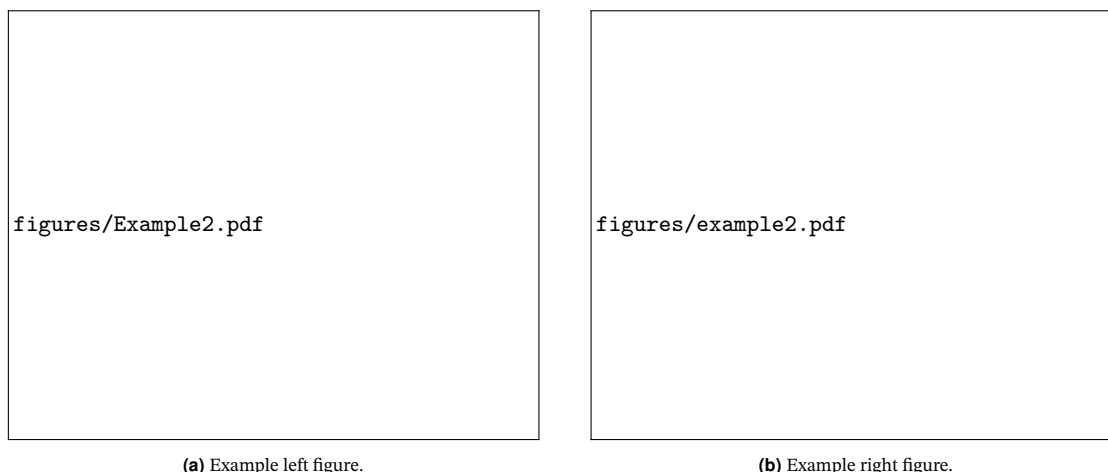


Figure 3. Example figure that covers the width of the page obtained from PGFPlots [4].

page. The space between the figures can also be changed using the `\hspace{Xpt}` command.

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5.3. Insert a table

In the same way as the figures, you can place tables in one or two columns, depending on the length of the table.

Table 1, shows an example table that covers the width of the page positioned at the bottom of a new page.

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6. Appendix

You may refer to an external document. Keep in mind to hand in this/these files.

■ References

- [1] Dau, H. A., et al. (2019). *The UCR Time Series Archive*. arXiv preprint arXiv:1810.07758. <https://arxiv.org/abs/1810.07758>

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Table 1. Table example that covers the width of the page.

Day	Min Temp	Max Temp	Summary
Monday	11°C	22°C	A clear day with lots of sunshine. Strong breeze will bring down the temperatures.
Tuesday	9°C	19°C	Cloudy with rain, across many northern regions.
Wednesday	10°C	21°C	Rain will still linger for the morning. Conditions will improve by early afternoon and continue throughout the evening.

Note: Obtained from L^AT_EX tables [2].